



Bluetooth light switch

Avi-on is the world's first wireless and movable switch that can also be controlled via a smartphone <http://dgit.in/Avi-on>



A real hoverboard

Arx Pax is trying to build a hoverboard using the same electromagnetic tech used in MagLev trains <http://dgit.in/HoverBrd>

Space Age

Mars. The purpose of the Mars Orbiter Mission, although stated to be a "technology demonstration", was, more meaningfully, about the scientific resilience that Indian scientists and researchers demonstrate, given their limited resources.

Powering MOM with Jugaad

The ISRO was only allocated ₹4,000 crore for its annual budget in 2013. Just to put that in context - the first successful American mission to Mars orbit cost ₹3,812 crore (adjusted for inflation). Indians scientists on the other hand had to use the whole ₹4,000 crore for dozens of projects across the board, not just one epic mission like the



MOM's solar panels

Americans. So how did they do it? Well, *jugaad* seems to be the operative word. In many a context, the word "*jugaad*" can appear to be a pejorative, but it's also the cornerstone of innovation.

The idea for pursuing a mission to Mars came in the wake of the launch of the Chandrayaan-1 in 2008. The mission concept and proposal was completed in 2010 with a thorough feasibility study conducted at a cost of ₹125 crore to the ISRO. The government approved the project in August 2012 as a worthy pursuit, intending to demonstrate India's advanced space-faring capabilities. The satellite design itself was calculated to be over ₹150 crore and it finally cost us ₹153 crore. Most of the other costs went into setting up ground-based tracking stations and support systems which would continue to have value for future mission as well.

An Adventure of Many Quests

From the moment of government approval the clock had started for ISRO and the

launch window of 28 October 2013 was decided upon. Since all spacefaring missions have to be conducted during certain "launch windows" when the rotation of the Earth around the Sun and the Earth's axis rotation make for an optimum trajectory. For Mars, this window occurs once every 780 days, a window where the least amount of energy is required to make the journey. A setback occurred almost immediately; adverse weather conditions affecting the ability to establish sea-based monitoring stations in the Pacific Ocean. A new launch date was then set - 5 November 2013 - less than 15 months from the date of the proposal being confirmed.

The mission was a technology demonstration but it needed to go above and beyond just the ability to reach Mars orbit. The spacecraft itself was designed to carry an instrumentation suite with which to conduct scientific readings of the planet. This payload would perform critical scientific functions once the launch vehicle had successfully attained a stable orbit around the planet. The instrumentation was integrated with the spacecraft in April 2013 and testing was scheduled to begin in August of the same year leaving very little margin for error due to the fast approaching launch window.

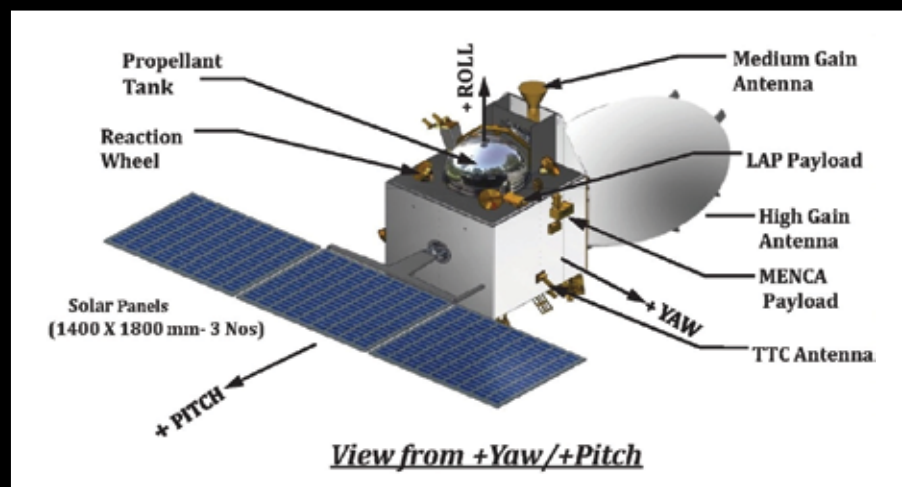
The initial plan called for the use of ISRO's geosynchronous Satellite Launch Vehicle (GSLV) but due to glitches in the cryogenic engines, there would be a three year delay in the launch. Not willing to wait ISRO decides that instead of devoting massive resources - of both time and money

- to a completely brand new launch system, it would be best to base the MOM spacecraft on the Chandrayaan-1 Moon Orbiter which was a Polar Satellite Launch Vehicle (PSLV). However this presented its own set of problems, the largest of which being the lower power of the PSLV compared to the GSLV. This is where one of the many *jugaad* decisions was taken. ISRO choose to forfeit a direct-to-Mars trajectory in favour of an iterative orbit, acceleration trajectory. A riskier venture but one that ISRO were confident in.

Ready Your Engines

With the same core spacecraft design and system structures from Chandrayaan-1 the scientists were certain that the mission objectives would be met. The power configuration of the spacecraft would effectively facilitate all the instruments mounted on the inner panels of the spacecraft. The cuboidal Mangalyaan craft, like its predecessor, used a composite and metallic honeycomb panel sandwich design for maximum flexural rigidity, allowing for lower weight constructions. This choice limited the launch mass to a svelte 1,337 (no, we're not joking) kilograms - with the craft weighing in at 475 kilograms, inclusive of payload, and 862 kilograms of fuel. These savings were a crucial part of the cost saving for the Mars Orbiter Mission.

The major source of power for launch was the bi-propellant Main Propulsion System and an Attitude Control System. The propulsion system consisted of two spherical tanks, each containing 390 liters



The indigenously designed system allows for a more compact configuration